

CostGuard

Cost-regression testing for AI agents

UiPath AgentHack. Build the AI agents of tomorrow.



CostGuard



Built solo by Leonardo Calancea · Track 3 (Test Cloud)

Problem statement and proposed solution

Problem

- Teams ship AI agents fast. A small prompt or model change can quietly triple token spend while still passing every correctness test.
- QA checks whether the agent is right, never what it costs to be right. The bills are board-level: FinOps X 2026 called it the Great Token Panic, and one 4-agent loop burned about \$47K in 11 days.
- There is a correctness gate before production. There is no cost gate.

• *Scores: Business Impact*

Solution

- CostGuard adds a first-class test type: cost-regression testing, run and governed on UiPath.
- Before a change ships, it runs the agent many times on a Test Cloud scenario set, measures cost per correctly-processed invoice (not per token), compares to the last approved baseline, and blocks a regression. The call goes to a human in Action Center.
- Cost is paired with a quality score, so a cheaper-but-worse agent cannot slip through. The line nobody else can draw: cost per business outcome.
- *Scores: Creativity · Platform Usage*

Benefits and technologies used

Details

End-user	Platform, QA and FinOps engineers shipping AI agents to production
User department	Engineering, QA, Platform Ops, FinOps
Industries	Any enterprise running production AI agents: finance, insurance, BPO, shared services
UiPath products used	Test Cloud (the gate), Coded Agents / Python SDK (engine + Cost Explainer), Orchestrator (serverless job), Action Center (HITL), Document Understanding (patient), AI Trust Layer, LLM Gateway

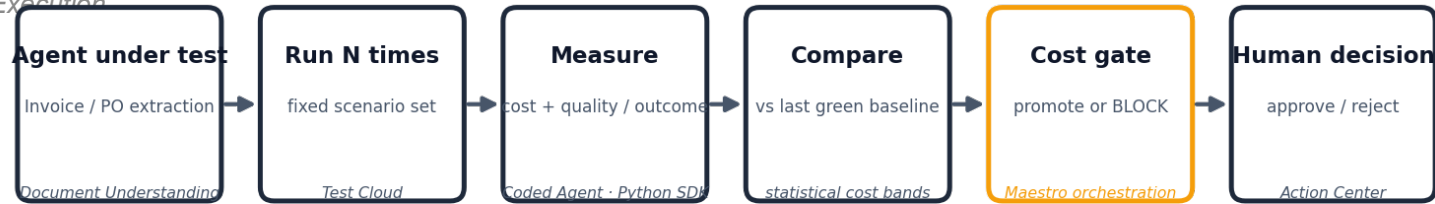
Benefits, impact and outcomes

- Blocks silent cost regressions before production. In the demo a “smarter” candidate scored +4.5% accuracy but cost 7× more per processed invoice, and CostGuard blocked it. On real UiPath-gateway models the same trap cost 13× for no accuracy gain.
- It prices agents in dollars per business outcome, works on any agent or framework, and a human always owns the final call.
- *Scores: Business Impact · Technical Execution*

Solution architecture

A change goes in. CostGuard runs the agent many times on UiPath, prices each correct outcome, compares to the baseline, and blocks a regression. A human owns any block.

Scores: Platform Usage · Technical Execution

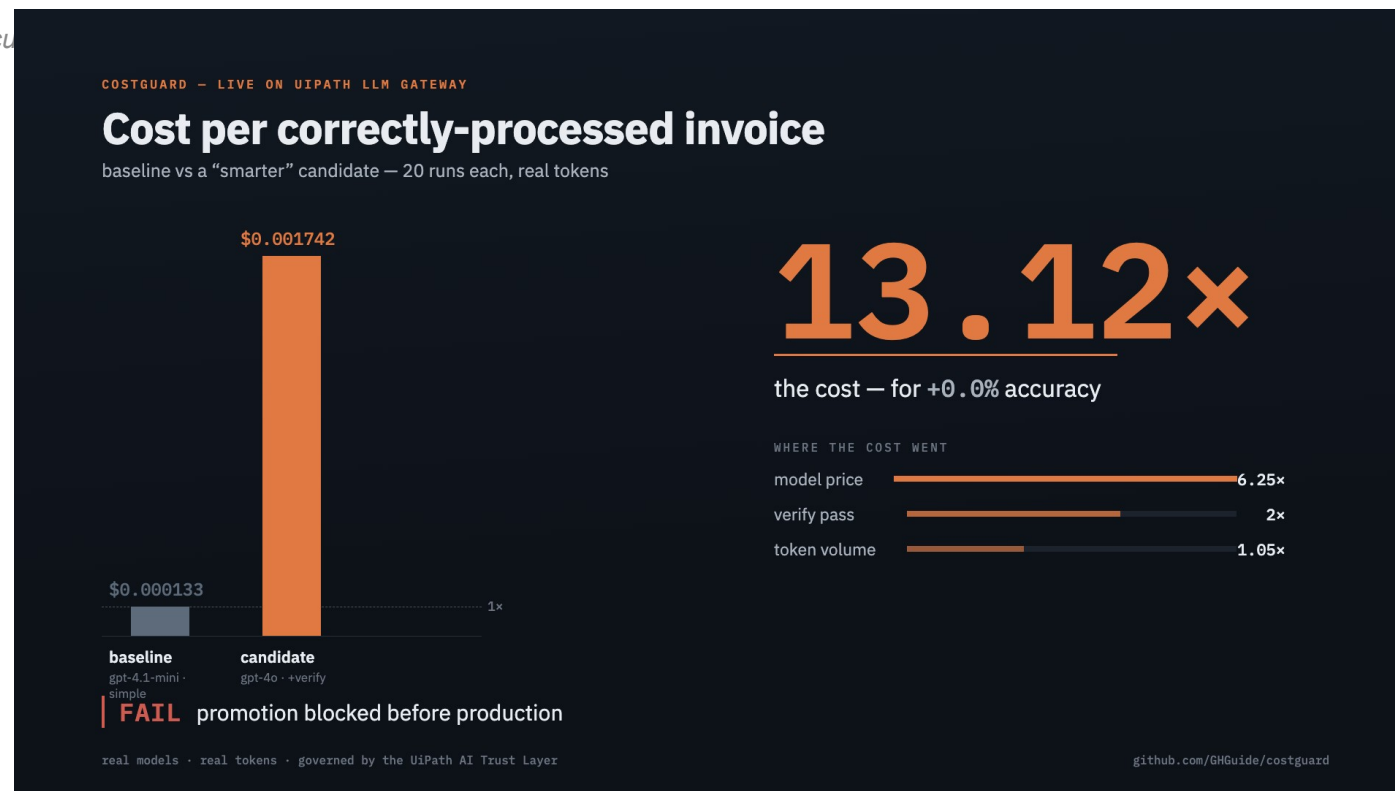


Live proof: real, not a slide

A live serverless job on UiPath Automation Cloud returned verdict FAIL: cost 7.26× for +4.5% accuracy. On the AI Trust Layer gateway the same upgrade cost 13.12× for zero gain.

The verdict is registered as a first-class Test Cloud result. The gate itself scores 30 out of 30 on a labelled eval.

Scores: Completeness · Technical Execution



THANK you.

Looked fine. Cost 7× more. Blocked before it shipped.

CostGuard · github.com/GHGuide/costguard